Schillings Air × TA-14

Evidence-Governed HVAC Decisions and Admissible Execution Boundaries

Technical Proposal V2.0 | November 2026 | Discussion draft

Proposed paid shadow-mode pilot for one bounded HVAC asset and decision class. This document is not evidence of an existing partnership, endorsement, deployment, measured savings, or an authorized control integration.

No admissible evidence. No admissible execution.

Prepared by TA-14 Authority for technical discussion. Any commercial terms, site selection, security approvals, control permissions and responsibilities require written agreement.

01 / EXECUTIVE DECISION BRIEF

Executive decision brief

The proposed collaboration preserves Schillings Air engineering, diagnostic expertise, field authority and existing building-control systems. TA-14 adds a distinct examination of evidence, applicable authority and established standing at the moment a consequential HVAC decision is proposed.

The first engagement is deliberately read-only: select one asset, one intervention class, one set of interfaces and an agreed set of adversarial scenarios. Replay proposed decisions against records without issuing commands to equipment. A later control phase is not included or implied.

The decision sought is permission to scope a paid discovery and shadow-mode pilot, not permission to actuate a device or represent a successful integration. The outcome is a traceable technical finding that can be accepted, revised or rejected by the parties.

Required evidence / deliverables

• Identify sponsor, technical owner and safety authority.

• Confirm pilot asset and precise decision class.

• Agree paid scope, schedule and data-access conditions.

02 / PROBLEM STATEMENT AND CONSEQUENCE BOUNDARY

Problem statement and consequence boundary

Building automation routinely combines observations, predictions, diagnostics and recommendations. A sensor can be accurate while its reading is stale, misattributed, incomplete for the proposed consequence or disconnected from the person authorized to approve the action.

The proposed TA-14 boundary separates a claim that action would be useful from a determination that action is permissible NOW. A valid communications connection, high confidence score or optimization objective does not itself confer execution authority.

An example request to increase outdoor air at AHU-3 is only a teaching scenario. It is not a recommendation to change actual ventilation rates. Engineering design constraints, equipment protections, local requirements and the existing safety chain retain precedence.

03 / PARTIES, RESPONSIBILITIES AND NON-CLAIMS

Parties, responsibilities and non-claims

Schillings Air, if it chooses to participate, would identify its engineering owner, applicable controls contractor, facility representative and operational constraints. TA-14 would document the independent examination logic, evidence contracts, determinations and replay outputs.

Neither party is assigned duties by this draft. No party is authorized to command equipment through the showroom, pilot documents or examples. Control ownership, professional responsibility, cybersecurity, privacy and insurance terms remain subjects for a separately executed agreement.

Public illustrations, specimen reports, synthetic telemetry and narrated demonstrations are educational only. They are not independently verified field records, site findings, signatures, certified hashes or actual client outcomes.

04 / ARCHITECTURE AND EIGHT-LINK CHAIN

Architecture and eight-link chain

The governing sequence is Reality → Record → Continuity → Admissibility → Binding → Commit → Execution → Outcome. Each link must be individually attributable and independently examinable. A downstream state may not silently repair an upstream deficiency.

Reality denotes the physical condition. Record is the attributed observation. Continuity checks whether provenance and validity survive change. Admissibility tests fitness for the particular consequence. Binding associates the exact proposed action with authority and standing. Commit revalidates the binding at the instant of authorization. Execution records bounded action; Outcome documents independent post-action evidence.

A HOLD at any necessary link prevents simulated authorization. Reassessment is required after material change; an earlier ALLOW is not a perpetual permit.

05 / SCOPE AND EXCLUSIONS

Scope and exclusions

Proposed scope: one named HVAC asset, one bounded consequence such as a hypothetical outdoor-air setpoint adjustment, a defined evidence window, identified data sources, authorized roles and an agreed replay dataset. The pilot is read-only and may use synthetic or de-identified records.

Excluded unless separately agreed: direct BAS/BMS commands, PLC or safety-controller changes, refrigerant handling, field intervention, cloud-to-control write access, replacement of licensed professional judgment, compliance certification and promised energy or IAQ savings.

Changes in site, equipment class, command type, control envelope or data ownership require change control. The pilot must not quietly expand from observation to operation.

Required evidence / deliverables

- Written asset and decision-class boundary.
- Explicit in-scope and out-of-scope interfaces.
- Named sponsor and escalation route.

06 / DATA AND EVENT INTERFACE CONTRACT

Data and event interface contract

Each candidate decision event should contain event_id, asset_id, observed_at, ingested_at, source_id, source_type, source_version, measured_value, unit, quality_flag, provenance_ref, continuity_status, proposed_action, proposed_value, requested_by, authority_ref, scope_ref, validity_window and correlation_id.

Each examination response should contain determination (ALLOW, HOLD, DENY or ESCALATE), reason_codes, examined_evidence_refs, evaluated_at, policy_version, boundary_version, required_next_action and expiry. Binding and commit events must preserve immutable correlation to the exact proposed action and its authority context.

Illustrative payload: event_id=EVT-0001; asset_id=AHU-3; proposed_action=outside_air_adjustment; source_id=BAS-CO2-03; observed_at=2026-11-01T12:00:00Z; determination=HOLD; reason_codes=[EVIDENCE_NOT_FROZEN]. Values are examples, not live records.

Required evidence / deliverables

- Schema, field definitions and versioning rules.
- Missing-field and invalid-unit behavior.
- Clock, time-zone and provenance assumptions.

07 / EVIDENCE ADMISSIBILITY AND CUSTODY

Evidence admissibility and custody

An observed value is not admitted solely because a dashboard displays it. The examination asks whether the record is attributable, timely, relevant, continuous, complete enough for this decision and available for independent review.

Source identity and calibration status, where applicable, must be distinguishable from confidence estimates. Records must disclose gaps, substitution, interpolation, manual override, sensor drift and known limitations. Missing information remains visibly missing.

Any evidence freeze must record the snapshot identity, source references, version and time boundary. Later data cannot be retroactively inserted into a frozen baseline without a new version and a disclosed re-examination.

08 / AUTHORITY, STANDING AND BINDING

Authority, standing and binding

Applicable authority must cover the exact proposed action, equipment, operating mode, location, time window and actor. Established standing requires the contextual conditions necessary for that authority to apply, including safety and organizational preconditions.

A person may be authorized to review a diagnostic record without being authorized to change a setpoint. A control system may be technically capable of writing a value without the institutional or operational permission to do so.

At binding, the decision identifies the requested consequence, controlling policy, evidence snapshot and actor. At commit, these are checked again; stale, revoked, conflicting or changed context fails closed.

09 / DETERMINATION SEMANTICS

Determination semantics

ALLOW: all specified simplified prerequisites currently pass for a bounded action; it is not a guarantee of performance or an enduring permission. HOLD: necessary evidence, continuity, standing or authority is missing, stale or unresolved.

DENY: the action is prohibited by a hard safety, policy or scope boundary. ESCALATE: conflicting authority, ambiguous responsibility or another defined exception requires accountable human review. Escalation is not an alternate route to silent permission.

Decision reason codes should be stable, machine-readable and paired with a plain-language explanation. The pilot should demonstrate that each failure mode produces the intended determination and next action.

10 / HVAC PERFORMANCE RECORD SEQUENCE

HVAC performance record sequence

The proposed workflow is HPR-BL baseline → NIRET gate → Declared Diagnostic Determination (DDD) → justified and authorized intervention → HPR-V post-intervention performance record, where applicable.

The baseline must be complete for the selected method and frozen before DDD is issued. Routine maintenance may warrant no intervention. The NIRET threshold governs whether specialist diagnostic escalation is justified; it is not a standing authorization to connect gauges or handle refrigerant.

The post-work record is separate from the baseline and must disclose what changed, who performed the work, what was actually measured and which claims remain unproven.

11 / FAILURE MODES AND FAIL-CLOSED BEHAVIOR

Failure modes and fail-closed behavior

Failure cases include stale timestamps, broken source identity, mismatched asset IDs, missing authority references, revoked credentials, contradictory commands, sensor discontinuity, unexpected operating mode, unsafe envelope, network outage and replayed or duplicated events.

For each case the parties should predefine expected determination, reason code, allowed next action, notification target and recovery condition. Missing or inconsistent inputs must not silently become ALLOW. A network failure must not bypass the local safety chain.

If a decision is ALLOW at examination but the underlying conditions change before commit, the prior determination must be invalidated and a fresh examination required. The shadow-mode demonstration must include this adversarial sequence.

12 / SECURITY AND OPERATIONAL SAFETY

Security and operational safety

The pilot should use least-privilege read-only credentials, isolated test endpoints where available, access logging, data minimization, documented retention and a designated incident contact. Secrets must not be embedded in demonstrations, source control or PDF appendices.

Existing building safety interlocks, fire/life-safety controls, OEM protections, supervisory control and licensed contractor obligations remain controlling. TA-14 must never be presented as a replacement for those systems.

Before any later control integration, a separate security and hazard review must approve command envelopes, rollback behavior, manual override, failure recovery, monitoring and emergency stop procedures.

13 / PILOT WORKPLAN AND MILESTONES

Pilot workplan and milestones

Phase 0 - paid discovery: confirm stakeholders, site permissions, scope, asset identity, use case, data dictionary, safety exclusions and acceptance tests. Phase 1 - interface mapping: document sample records and event schemas, data lineage and authority roles.

Phase 2 - shadow replay: process agreed synthetic or authorized historical events, exercise normal and adversarial scenarios, and record decision outcomes without physical commands. Phase 3 - findings: deliver traceability, exceptions, acceptance matrix and prioritized integration recommendations.

Timing, fees, resource allocations and dependencies are to be agreed in a statement of work. No fixed schedule or price is represented as accepted in this draft.

14 / ACCEPTANCE CRITERIA AND TEST MATRIX

Acceptance criteria and test matrix

The parties should approve a test catalog before execution. The minimum set includes missing evidence→HOLD, stale evidence→HOLD, missing authority→HOLD, conflicting authority→ESCALATE, prohibited safety condition→DENY, valid bounded scenario→ALLOW and changed context before commit→blocked.

Every test should capture input event, evidence references, expected and actual determination, reason codes, evaluated time, policy version, operator, and pass/fail rationale. Repeatability requires the same versioned inputs and policy configuration.

Acceptance is based on observed reproducibility and documented exceptions, not on an attractive dashboard, synthetic performance improvement or an unverified claim of production readiness.

15 / DELIVERABLES AND OWNER REVIEW

Deliverables and owner review

Deliverables proposed: agreed scope register; source and event interface contract; evidence admissibility matrix; authority-role matrix; versioned decision logic description; shadow-mode replay log; failure-mode results; acceptance test report; and a final recommendations memo.

An owner-facing report should distinguish observed reality, source record, admissibility finding, authorized scope, action status and post-action outcome. Illustrative signatures, hashes and verification marks must be identified as specimens until independently validated.

The report should disclose unresolved issues, data gaps, test limitations and all reasons why the evidence does not support a stronger claim.

16 / COMMERCIAL FRAMEWORK AND DECISION GATES

Commercial framework and decision gates

This is a proposed paid engagement, not an offer at a stated price. A statement of work should establish discovery fee, milestone payments, authorized expenses, responsibilities, acceptance procedure, confidentiality, intellectual-property treatment, termination, liability and use of names or logos.

Gate A authorizes paid discovery only. Gate B approves the bounded read-only shadow pilot after technical and data-access review. Gate C accepts or rejects the findings. Gate D, if ever proposed, is a separately negotiated and independently approved control integration.

Neither publication of this document nor use of the interactive showroom constitutes Gate A approval or permission to conduct field work.

17 / OPEN DECISIONS AND REQUESTED RESPONSE

Open decisions and requested response

To proceed, the parties would need to name a technical sponsor, identify one candidate asset and intervention class, determine whether representative records can be shared, and identify who has authority over site access, cybersecurity and safety.

Proposed first meeting agenda: 15 minutes for use-case definition, 15 minutes for interface and evidence availability, 15 minutes for authority and safety boundaries, and 15 minutes for commercial scope and next steps.

The requested response is whether Schillings Air wishes to discuss a paid, bounded shadow-mode feasibility engagement. A negative or deferred answer closes the inquiry without any implied obligation.

18 / APPENDIX: EXAMPLE REPLAY AND GLOSSARY

Appendix: example replay and glossary

Illustrative replay: T0 request received, missing source continuity→HOLD. T1 continuity and authority established→ALLOW for a narrowly defined simulated action. T2 a fresh condition change arrives before commit→earlier ALLOW invalidated. T3 commit attempt→BLOCKED/HOLD. No physical command is issued.

Glossary: HPR-BL = baseline performance record; NIRET = threshold for specialist diagnostic escalation; DDD = Declared Diagnostic Determination; HPR-V = post-intervention performance record; binding = association of exact proposed consequence with evidence, authority and standing; commit = final revalidation before consequential execution.

Limitations: this draft has not been approved by Schillings Air and does not contain a verified field dataset, signed commercial terms, final schema, hazard analysis or completed acceptance test. The accompanying showroom is an educational concept.